

DOMAIN 2: AI for National Technology Solutions

1) Problem Statement: Terrain Aided Navigation (TAN) System

Modern airborne platforms, particularly cruise missiles and autonomous vehicles, rely heavily on GPS for accurate position and guidance updates. However, in contested or hostile environments, GPS signals are highly vulnerable to jamming and spoofing, which can severely degrade or completely deny positioning accuracy — compromising mission success and vehicle safety. There is, therefore, a critical need for a robust, GPS-independent navigation solution that can maintain accurate positioning even in denied or degraded signal environments.

Terrain Aided Navigation (TAN) addresses this gap by determining an airborne vehicle's position relative to the terrain it is flying over. The system uses onboard sensors — including a barometer and radio altimeter — to observe terrain features along the flight path in real time. These observed features are compared against pre-stored, referenced terrain elevation data to estimate the vehicle's position and altitude error. In the event of GPS jamming or spoofing, this terrain-derived position estimate can be used to correct and update the vehicle's mid-course guidance system, ensuring continued navigational accuracy without dependence on satellite signals.

The core research challenge lies in designing a TAN algorithm capable of reliably localizing the vehicle using terrain contour data across diverse and natural terrain types, while also optimizing the mission flight plan to maximize the availability and accuracy of TAN system outputs throughout the flight envelope.

Objectives:

1. Design and develop a TAN algorithm for vehicle localization using terrain contour matching.
2. Optimize mission planning to maximize TAN system output availability across varied terrain profiles.

Expected Deliverables:

- Detailed design, model, and source code of the TAN algorithm.
- Analysis reports and test results validating TAN performance across various natural terrain types.

Application: Cruise missile navigation in GPS-denied environments.

2) Problem Statement: Scene Aided Navigation (SAN) System

Airborne platforms such as cruise missiles and UAVs conventionally depend on GPS for accurate position updates during flight. In operational environments where GPS signals may be denied or spoofed by adversarial systems, the vehicle's guidance system loses its primary means of position correction, directly threatening mission accuracy and success. This makes it essential to develop an alternative, vision-based navigation capability that functions independently of satellite signals.

Scene Aided Navigation (SAN) offers a solution by estimating an airborne vehicle's position relative to the imaged scene beneath it, using Electro-Optical (EO — Visible & IR) camera sensors. Prior to flight, the system is loaded with processed satellite imagery/maps of the expected flight corridor. During flight, the SAN system captures a sequence of short-exposure frames from low altitude, nominally along the nadir direction, using a video camera. Each captured frame is matched against the onboard stored map to estimate the sensor's (and hence the vehicle's) position. This scene-matched position estimate can then be used to update an INS (Inertial Navigation System) based guidance system in the event GPS is denied or spoofed.

The central research challenge is to design a SAN algorithm capable of robust and accurate scene matching for vehicle localization across diverse imaged scenes and flight conditions, and to model and simulate suitable sensor configurations for integrating SAN across different classes of air vehicles.

Objectives:

1. Design and development of the SAN algorithm for vehicle localization based on scene matching.
2. Modelling and simulation of sensor configurations for SAN system integration across various air vehicle platforms.

Expected Deliverables:

- Detailed design, model, and source code of the SAN algorithm.
- Analysis reports and test results of SAN algorithm performance across varied scene types.
- SAN sensor configurations tailored for cruise missiles and UAVs.

Applications: Cruise missile and UAV navigation in GPS-denied environments.

3) Problem Statement: Disposal of Discarded High-Performance Fibers (HPFs)

High-performance fibers (HPFs) — including aramid (p-aramid, m-aramid), ultra-high molecular weight polyethylene (UHMWPE), and carbon fibers — are extensively used across aerospace, defense, automotive, and protective textile applications owing to their exceptional strength-to-weight ratio, chemical resistance, and thermal stability. However, these very properties that make HPFs valuable in service also make them extremely difficult to manage at end-of-life. HPF-based composites and textiles are highly resistant to degradation, thermally stable, and largely incompatible with conventional recycling or landfill disposal methods, resulting in significant resource wastage and ecological burden.

Current disposal practices for such materials — primarily incineration or landfilling — are energy-intensive, environmentally unsustainable, and fail to recover the substantial material value embedded in these high-performance fibers. As defense and allied sectors generate increasing volumes of discarded HPF-based composites, protective gear, and components, there is an urgent need to develop disposal and recycling methods that are sustainable, economically viable, and scalable, enabling a shift toward circular economy practices in defense waste management.

The core research challenge lies in developing eco-efficient and technically feasible strategies to disassemble, recover, and repurpose discarded HPFs, while ensuring the process is both environmentally sound and economically competitive against existing disposal routes.

Objective:

To develop and evaluate eco-efficient and technically feasible strategies for the disposal, recycling, and recovery of discarded high-performance fibers, aimed at minimizing environmental impact while maximizing material value recovery.

Expected Deliverables:

- A comprehensive database of degradation and recyclability characteristics of major HPFs.
- Optimized disposal/recycling protocols specific to each fiber type (e.g., solvent-based recovery, pyrolysis parameters).
- Comparative environmental and economic performance metrics benchmarked against existing disposal routes.
- Prototype demonstration of recycled HPF-based materials or composites.
- Technical guidelines for sustainable HPF waste management.

Applications:

- Defense and aerospace sectors, for recycling obsolete composites and protective gear.
- Automotive industry, for recovery of fibers from end-of-life components.
- Protective textiles, for closed-loop recycling systems.

4) Problem Statement: Carbon Credit Marketplace & MRV Platform

India's Carbon Credit Trading Scheme (CCTS) — the country's first mandatory domestic emissions trading system — is transitioning industrial entities from the erstwhile Perform, Achieve and Trade (PAT) scheme into a legally binding, intensity-based carbon market. Compliance obligations are already in force for roughly 490 industrial entities across seven energy-intensive sectors (aluminium, cement, chlor-alkali, pulp and paper, petroleum refining, petrochemicals, and textiles), with iron & steel and fertilizers expected to bring the total to nine sectors once finalized. These nine sectors together account for approximately 16% of India's total GHG emissions, and once trading begins, CCTS will cover over 700 million tonnes of CO₂e — placing it among the largest emissions trading systems globally. [Carbonmarketnetwork](#)

Despite this scale, there is currently no unified, transparent digital infrastructure for the full lifecycle of carbon credit management — spanning emissions monitoring, automated reporting, third-party verification, credit issuance, and exchange-based trading. Obligated entities face fragmented, manual, and audit-heavy MRV (Monitoring, Reporting, and Verification) processes, increasing compliance costs and the risk of reporting errors, while the absence of a trusted trading layer limits price discovery and liquidity in the emerging Carbon Credit Certificate (CCC) market.

Solution:

A blockchain-based carbon credit marketplace that combines automated MRV workflows with verifiable, tamper-resistant credit issuance and a transparent trading exchange — reducing compliance friction for obligated entities while enabling credible price discovery for Carbon Credit Certificates.

Objectives:

1. Automate emissions data collection, reporting, and third-party verification (MRV) for CCTS-obligated entities.
2. Build a blockchain-anchored registry and trading layer for issuance, transfer, and settlement of Carbon Credit Certificates.

Expected Deliverables:

- MRV automation module integrated with plant-level emissions data sources.
- Blockchain-based credit registry with immutable audit trail.
- Trading/exchange interface for CCC price discovery and settlement.
- SaaS compliance dashboard for obligated entities and verifiers.

5) Problem Statement: AI-Powered Automated Digitization of Orthophotos for GIS and LULC Mapping

High-resolution drone and satellite orthophotos have become foundational data sources for urban planning, agriculture, forestry, infrastructure development, and environmental monitoring. However, converting these raster images into usable, structured geospatial information — such as road networks, building footprints, farm boundaries, and land-use classifications — still relies heavily on manual digitization by GIS analysts. This process is labor-intensive, time-consuming, and expensive, creating a bottleneck that limits how quickly and how frequently large geographic areas can be mapped and updated, particularly as drone and satellite imagery volumes continue to grow.

There is a clear need for an AI/ML-driven solution that can automatically interpret orthophotos and related geospatial datasets, extracting accurate, GIS-ready vector data at scale — reducing turnaround time and human effort while maintaining the positional and classification accuracy required for planning and monitoring applications.

Objectives:

1. Design and develop an AI/ML platform capable of automatically detecting, classifying, counting, and digitizing land-use and infrastructure features from orthophotos and related geospatial inputs.
2. Enable export of extracted features into industry-standard GIS formats with associated confidence scoring and a quality-control workflow to support human validation where needed.

Expected Capabilities:

- Road detection and centerline extraction
- Building footprint extraction
- Farm boundary delineation
- Tree detection and counting
- Water body extraction
- LULC (Land Use/Land Cover) classification
- Export to Shapefile, GeoJSON, GeoPackage, and FileGDB
- Confidence scoring and QC workflow for extracted features

Input Datasets:

- Drone orthophotos
- Satellite imagery
- DSM/DTM (Digital Surface Model / Digital Terrain Model)
- DEM (Digital Elevation Model)
- LiDAR data (optional)
- Existing GIS layers (for reference/validation)

Expected Deliverables:

- Trained AI/ML model(s) for feature extraction
- GIS-ready vector datasets
- LULC classification map
- Tree inventory dataset
- Road network dataset
- Building footprint dataset
- Farm boundary dataset
- Accuracy/validation report